Full Stack Web Development

TCS 693

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Full Stack Web Development TCS 693

Sl. No.	Contents	Contact Hours
1	**Unit 1:** **HTML** Basics of HTML, formatting and fonts, commenting code, color, hyperlink, lists, tables, images, forms, XHTML, Meta tags, Character entities, frames and frame sets, Browser architecture and Web site structure. Overview and features of HTML5. **CSS** Need for CSS, introduction to CSS, basic syntax and structure, using CSS, type of CSS, background images, colors and properties, manipulating texts, using fonts, borders and boxes, margins, padding lists, positioning using CSS, Introduction to Bootstrap.	8
2	**Unit 2:** JavaScript: Client-side scripting with JavaScript, variables, functions, conditions, loops and repetition, Pop up boxes. Advance JavaScript: JavaScript and objects, JavaScript own objects, the DOM and web browser environments, Manipulation using DOM, forms and validations, JSON.	8

3	**Unit 3:** **XML**: Introduction to XML, uses of XML, simple XML, XML key components, DTD and Schemas. **Ajax** : Introduction to Ajax , XMLHttpRequest Methods and Properties, JavaScript code for Ajax , Implementing Ajax techniques with a server scripting language , Handling the Response, Ajax with JSon **JQuery**: jQuery Introduction, Install and Use jQuery Library, jQuery Syntax, Ajax with jQuery, Load method, jQuery get and getJson methods.	10
4	**Unit 4:** **PHP** XAMPP Server Configuration, Introduction and basic syntax of PHP, decision and looping with examples, PHP and HTML, Arrays, Functions, Browser control and detection, string, Form processing, Files. Advance Features: Cookies and Sessions, Basic commands with PHP examples, Connection to server, creating database, selecting a database, listing database, listing table names, creating a table, inserting data, altering tables, queries, deleting database, deleting data and tables.	10

5	**Unit 5:** MERN Web Application Deployment, Content Management System (CMS). MERN Stack: MongoDB: Overview , Environment, Data Modelling, Database Operations. Express: Installing ExpressJS, Environment, Routing React: React Intro, , React Lifecycle, Building Forms using React, states and components. Node: Install node, simple server, HTML and JSON Response.	12
	Total	**48**

World Wide Web (WWW)?

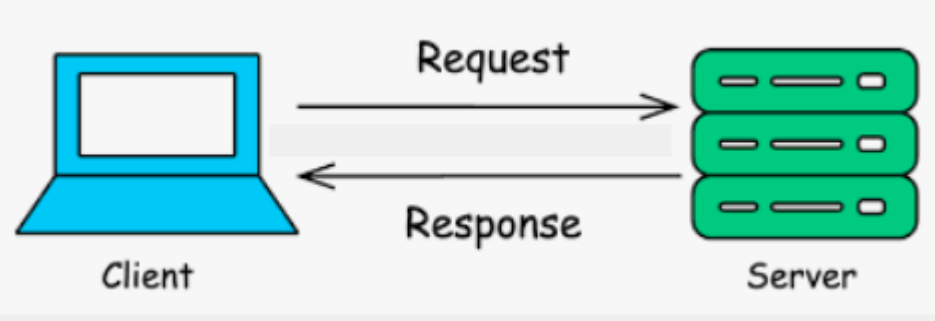
The **World Wide Web (WWW)** is a system of interlinked web pages that are accessed through the Internet using a **web browser**.

- Invented by Tim Berners-Lee
- Uses HTTP/HTTPS protocol
- Information is stored in the form of web pages
- Web pages are written using HTML

Client–Server Architecture

How a Website Works

- User enters URL in browser
- Browser sends request to server
- Server processes request
- Response sent back to browser
- Browser renders web page



Components

- Client: Browser (Chrome, Edge)
- Server: Apache, Nginx
- Protocol: HTTP / HTTPS

Website Structure

Frontend (Client Side)

- HTML → Structure
- CSS → Styling
- JavaScript → Behavio

Backend (Server Side)

- PHP, Node.js, Python
- Database (MySQL, MongoDB)

```
<!doctype html>
<html>
<head>
  <title>A web page</title>
</head>
<body>
  <h1>This web page is great!</h1>
  <p>This is a test web page for
  that serves as an example for this
  lecture.</p>
</body>
</html>
```

Tag

Close Tag

Text Content

Whitespace insignificant

HTML

HTML (HyperText Markup Language) is a markup language used to structure web pages.

Features

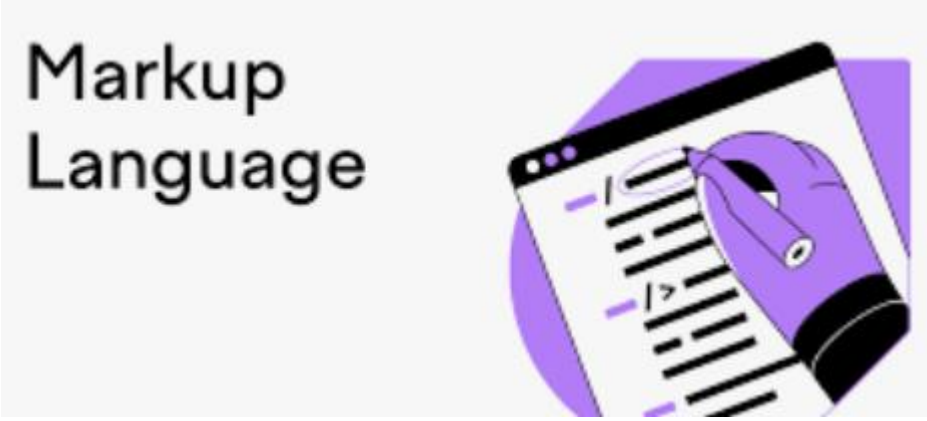
- Easy to learn
- Platform independent
- Supports multimedia
- Not a programming language

Markup Language

A markup language is used to structure, format, and present data. It does not perform logic, calculations, or decision-making.

Key Characteristics

- Describes structure and layout
- No variables, loops, or conditions
- Interpreted by browsers or parsers
- Used for content presentation



Examples

- HTML – Web page structure
- XML – Data storage and transfer
- SGML – Standard generalized markup
- XHTML – HTML with XML rules
- Markdown – Lightweight formatting

Scripting Language

A scripting language is used to automate tasks, add interactivity, and control behavior of applications—often inside another environment.

Key Characteristics

- Interpreted (no compilation step)
- Used for **client-side or server-side tasks**
- Supports variables, loops, conditions
- Executes line-by-line

Examples

- **JavaScript** – Client-side scripting
- **PHP** – Server-side scripting
- **Python** – Automation & scripting
- **Ruby** – Web scripting
- **Shell Script** – OS automation

Programming Language

A programming language is used to develop complete software applications, perform complex computations, and build systems from scratch.

Key Characteristics

Can be compiled or interpreted

Supports **OOP, data structures, algorithms**

Used to build **desktop, mobile, web, and system software**

High performance and scalability

Examples

C – System programming

C++ – High-performance apps

Java – Enterprise & Android apps

Python – General-purpose programming

C# – Windows & game development

Feature	Markup Language	Scripting Language	Programming Language
Purpose	Structure & formatting	Automation & interaction	Full application development
Logic Support	❌ No	✅ Yes	✅ Yes
Compilation	❌ No	❌ No	✅ Yes (mostly)
Execution	By browser/parser	Interpreter	Compiler/Interpreter
Complexity	Low	Medium	High
Performance	Not applicable	Moderate	High
Examples	HTML, XML	JavaScript, PHP	C, C++, Java

- **Markup language** defines *what the content is*.
- **Scripting language** defines *how the content behaves*.
- **Programming language** defines *how the system works*.

First name:

Last name:

Gender : Male Female

Email:

Date of Birth: dd/mm/yyyy

Address :

Submit

```
<form>
   <label>First name:</label>
   <input type="text" name="firstname"><br><br>
   <label>Last name:</label>
   <input type="text" name="lastname"><br><br>
   <label>Gender :</label>
   <input type="radio" name="gender" value="Male"> Male
   <input type="radio" name="gender" value="Female"> Female
   <br><br>
   <label>Email:</label>
   <input type="email" name="email"><br><br>
   <label>Date of Birth:</label>
   <input type="date" name="dob"><br><br>
   <label>Address :</label><br>
   <textarea name="address" rows="4" cols="40"></textarea>
   <br><br>
   <input type="submit" value="Submit">
</form>
```

```
<form id="myForm" onsubmit="displayData(); return false;">
   <label>First name:</label>
   <input type="text" id="firstname"><br><br>
   <label>Last name:</label>
   <input type="text" id="lastname"><br><br>
   <label>Gender :</label>
   <input type="radio" name="gender" value="Male"> Male
   <input type="radio" name="gender" value="Female"> Female
   <br><br>
   <label>Email:</label>
   <input type="email" id="email"><br><br>
   <label>Date of Birth:</label>
   <input type="date" id="dob"><br><br>
   <label>Address :</label><br>
   <textarea id="address" rows="4" cols="40"></textarea>
   <br><br>
   <input type="submit" value="Submit">
</form>
```

```
<hr>
<h3>Entered Information</h3>
<div id="output"></div>
<script>
function displayData() {
    var fname = document.getElementById("firstname").value;
    var lname = document.getElementById("lastname").value;
    var email = document.getElementById("email").value;
    var dob = document.getElementById("dob").value;
    var address = document.getElementById("address").value;
    var gender = "";
    var genders = document.getElementsByName("gender");
    for (var i = 0; i < genders.length; i++) {
        if (genders[i].checked) {
            gender = genders[i].value;
        }
    }
```

```
document.getElementById("output").innerHTML =
        "<p><b>First Name:</b> " + fname + "</p>" +
        "<p><b>Last Name:</b> " + lname + "</p>" +
        "<p><b>Gender:</b> " + gender + "</p>" +
        "<p><b>Email:</b> " + email + "</p>" +
        "<p><b>Date of Birth:</b> " + dob + "</p>" +
        "<p><b>Address:</b> " + address + "</p>";
}
</script>
</body>
</html>
```

Registration Form

First name: amit

Last name: singh

Gender : Male Female

Email: xyz@gmail.com

Date of Birth: 01/01/2026

Address :

GEU

Submit

Entered Information

First Name: amit

Last Name: singh

Gender: Male

Email: xyz@gmail.com

Date of Birth: 2026-01-01

Address: GEU

Check Boxes

```
<html>
<head>    <title>Checkbox Example</title> </head>
<body>
<h2>Select Your Skills</h2>
<form onsubmit="showSkills(); return false;">
   <input type="checkbox" id="html" name="skills" value="HTML">
   <label for="html">HTML</label><br>
   <input type="checkbox" id="css" name="skills" value="CSS">
   <label for="css">CSS</label><br>
   <input type="checkbox" id="js" name="skills" value="JavaScript">
   <label for="js">JavaScript</label><br><br>
   <input type="submit" value="Submit">
</form>
<hr><h3>Selected Skills:</h3>
<div id="output"></div>
```

```
<script>
function showSkills() {
   var skills = document.getElementsByName("skills");
   var selectedSkills = [];

   for (var i = 0; i < skills.length; i++) {
      if (skills[i].checked) {
         selectedSkills.push(skills[i].value);
      }
  }
   if (selectedSkills.length === 0) {
      document.getElementById("output").innerHTML =
         "<p>No skill selected</p>";
   } else {
      document.getElementById("output").innerHTML =
         "<p><b>You selected:</b> " + selectedSkills.join(", ") + "</p>";
   }
}
</script>
</body>
</html>
```

Output

Select Your Skills

- [x] HTML
- [x] CSS
- [ ] JavaScript

Submit

Selected Skills:

You selected: HTML, CSS

Dropdown

```
<html>
<head>    <title>Dropdown Example</title> </head>
<body> <h2>Select Your Course</h2>
<form onsubmit="showCourse(); return false;">
   <label for="course">Course:</label>
   <select id="course">
      <option value="">-- Select Course --</option>
      <option value="B.Tech">B.Tech</option>
      <option value="M.Tech">M.Tech</option>
      <option value="BCA">BCA</option>
      <option value="MCA">MCA</option>
   </select>    <br><br>
   <input type="submit" value="Submit">
</form>
```

```
<h3>Selected Course:</h3>
<div id="output"></div>
<script>
function showCourse() {
   var course = document.getElementById("course").value;

   if (course === "") {
      document.getElementById("output").innerHTML =
         "<p>No course selected</p>";
   } else {
      document.getElementById("output").innerHTML =
         "<p><b>You selected:</b> " + course + "</p>";
   }
}
</script>
</body>
</html>
```

Select Your Course

Course: M.Tech

Submit

Selected Course:

You selected: M.Tech

Popup boxes

Alert

- Displays a **simple message**
- No user input
- Only **OK** button

```
<html><body>
<h2>Alert Box Example</h2>
<button onclick="showAlert()">Click Me</button>
<script>
function showAlert() {
    alert("Form submitted successfully!");
}
</script>
</body></html>
```

Confirm Box

- Takes Yes / No decision
- Returns true or false

```
<html> <body><h2>Confirm Box Example</h2>
<button onclick="showConfirm()">Delete Record</button>
<p id="result"></p>
<script>
function showConfirm() {
   var choice = confirm("Are you sure you want to delete?");

   if (choice) {
      document.getElementById("result").innerHTML = "Record deleted";
   } else {
      document.getElementById("result").innerHTML = "Action cancelled";
   }
}
</script> </body> </html>
```

Prompt Box

- Takes user input
- Returns entered value or null

```
<html> <body><h2>Prompt Box Example</h2>
<button onclick="showPrompt()">Enter Name</button>
<p id="output"></p>
<script>
function showPrompt() {
   var name = prompt("Enter your name:");

   if (name === null || name === "") {
      document.getElementById("output").innerHTML = "No name entered";
   } else {
      document.getElementById("output").innerHTML =
         "Hello, " + name;
   }
}
</script> </body> </html>
```

Basic Form Validation

Form validation ensures that user input is correct and complete before submitting data to the server.

It can be done using:

- HTML5 attributes (client-side)
- JavaScript logic (custom client-side)

Validation improves:

- Data accuracy
- User experience

```
<form onsubmit="return validateForm()">
    Name: <input type="text" id="name"><br><br>
    <input type="submit" value="Submit">
</form>
<script>
function validateForm() {
    var name = document.getElementById("name").value;

    if (name === "") {
        alert("Name is required");
        return false;
    }
    return true;
}
</script>
```

Required Fields

A required field must be filled before form submission.

<u>HTML5 Method</u>

```
<form>
   Name: <input type="text" required><br><br>
   <input type="submit">
</form>
```

JavaScript Method

```
<input type="text" id="fname">
<button onclick="check()">Submit</button>
<script>
function check() {
    if (document.getElementById("fname").value === "") {
        alert("Field cannot be empty");
    }
}
</script>
```

Email & Number Validation

Using HTML5

```
<input type="email" required>
```

Using JavaScript

```
<input type="text" id="email">
<button onclick="validateEmail()">Check</button>

<script>
function validateEmail() {
    var email = document.getElementById("email").value;

    if (!email.includes("@")) {
        alert("Invalid email");
    }
}
</script>
```

Number Validation

Using HTML5

```
<input type="number" min="1" max="100" required>
```

Using JavaScript

```
<input type="text" id="age">
<button onclick="validateAge()">Check</button>
<script>
function validateAge() {
    var age = document.getElementById("age").value;
    if (isNaN(age)) {
        alert("Enter a valid number");
    }
}
</script>
```

Pattern Attribute

The pattern attribute uses regular expressions to restrict input format.

Mobile Number

```
<input type="text"  pattern="[0-9]{10}"  title="Enter 10 digit mobile number“
required>
```

Password

```
<input type="password"
    pattern="(?=.*[A-Z])(?=.*[0-9]).{6,}"
    title="Must contain uppercase letter and number">
```

Example

```
<form onsubmit="return validate()">
   Email:
   <input type="email" id="email" required><br><br>
   Age:
   <input type="number" id="age" required><br><br>
   <input type="submit">
</form>
<script>
function validate() {
   var age = document.getElementById("age").value;
   if (age < 18) {
      alert("Age must be 18 or above");
      return false;
   }
   return true;
}
</script>
```

What are HTML5 Input Types?

HTML5 input types provide built-in validation, better UI controls, and improved user experience by restricting the type of data a user can enter.

```
<input type="text" placeholder="Enter your name">
<input type="password" placeholder="Enter password">
//Hides characters for sensitive data.
<input type="email" placeholder="Enter email" required>
//Validates email format automatically.
<input type="number" min="1" max="100" placeholder="Enter age">
//Accepts only numeric values.
<input type="date"> //Provides a calendar for date selection.
<input type="time"> //Allows selection of time.
<input type="color"> //Opens a color picker.
<input type="range" min="0" max="100">//Creates a slider control for numeric range.
<input type="file">//Allows file upload.
```

Variables & Data Types

Variables store data values.

JavaScript has dynamic typing (type can change).

Main data types: Number, String, Boolean, Undefined, Null, BigInt, Symbol, Object.

```
<script>
let a = 10;          // Number
let b = "GEU";       // String
let c = true;        // Boolean
let d;               // Undefined
let e = null;        // Null
console.log(typeof a, typeof b, typeof c, typeof d, e);
</script>
```

var, let, const

var → function-scoped, can be re-declared (older, avoid)

let → block-scoped, can be reassigned

const → block-scoped, cannot be reassigned (value fixed)

```
<script>
var x = 5;
var x = 10; // allowed (re-declare)
let y = 5;
// let y = 10; // not allowed (re-declare)
y = 10; // allowed (reassign)
const z = 5; // z = 10; // not allowed (reassign)
</script>
```

Primitive vs Non-Primitive Types

Primitive: stored by value (copy)

- Number, String, Boolean, Undefined, Null, BigInt, Symbol

Non-primitive: stored by reference

- Object, Array, Function

```
<script>
// Primitive (copy)
let p1 = 10;
let p2 = p1;
p2 = 20;
console.log(p1, p2); // 10 20

// Non-primitive (reference)
let arr1 = [1,2];
let arr2 = arr1;
arr2.push(3);
console.log(arr1, arr2); // [1,2,3] [1,2,3]
</script>
```

Operators

Arithmetic + - * / % ** ++ --

```
<script>
let a = 10, b = 3;
console.log(a+b, a-b, a*b, a/b, a%b, a**b);
</script>
```

Relational (Comparison)> < >= <= == != === !==

```
<script>
console.log(10 > 5);   // true
console.log(10 <= 5);  // false
</script>
```

Logical && || !

```
<script>
let age = 20;
console.log(age >= 18 && age <= 60); // true
console.log(!(age < 18));            // true
</script>
```

== vs ===

== compares values only (type conversion happens)

=== compares value + type (strict)

```
<script>
console.log(5 == "5");   // true (type conversion)
console.log(5 === "5");  // false (number vs string)
</script>
```

Switch

```
<script>
let day = 3;
switch(day) {
  case 1: console.log("Monday"); break;
  case 2: console.log("Tuesday"); break;
  case 3: console.log("Wednesday"); break;
  default: console.log("Invalid day");
}
</script>
```

Loops

For

```
<script>
for (let i = 1; i <= 5; i++) {
  console.log(i);
} </script>
```

While

```
<script>
let i = 1;
while (i <= 5) {
  console.log(i);
  i++;
}</script>
```

do-while

```
<script>
let j = 1;
do {
  console.log(j);
  j++;
} while (j <= 5);
</script>
```

Loop with Form Inputs

```
<!DOCTYPE html>
<html>
<body>

<h3>Select Subjects</h3>

<form onsubmit="showSubjects(); return false;">
  <input type="checkbox" name="sub" value="DBMS"> DBMS <br>
  <input type="checkbox" name="sub" value="OS"> OS <br>
  <input type="checkbox" name="sub" value="CN"> CN <br><br>
  <input type="submit" value="Submit">
</form>

<p id="out"></p>
```

```
<script>
function showSubjects() {
  let boxes = document.getElementsByName("sub");
  let selected = [];

  for (let i = 0; i < boxes.length; i++) {
    if (boxes[i].checked) {
      selected.push(boxes[i].value);
    }
  }
  document.getElementById("out").innerHTML =
    (selected.length === 0) ? "No subject selected" : "Selected: " + selected.join(", ");
}
</script>
</body>
</html>
```

Document Object Model (DOM) – Window Object

The Document Object Model (DOM) represents an HTML page as a structured tree of objects. It allows JavaScript to access, manipulate, and control web page elements dynamically. At the top level of the DOM hierarchy is the window object`, which represents the browser window or tab. All global JavaScript objects, functions, and variables automatically becom

Window Object

The window object is the global object in the browser environment. It controls browser-related features such as alerts, navigation, timing, screen size, and history.

- Parent of all other objects (document, navigator, screen, history)
- Represents the browser window
- Provides methods for interaction with the browse members of the window object.

Common Window Object Properties

Property	Description
window.document	Refers to the HTML document
window.location	Current URL of the page
window.history	Browser history
window.navigator	Browser information
window.screen	Screen resolution info
window.innerWidth	Width of browser window
window.innerHeight	Height of browser window

alert() – Display Message Box

```
<script>
  window.alert("Welcome to DOM!");
</script>
```

confirm() – Confirmation Dialog

```
<script>
  let result = window.confirm("Do you want to continue?");
  if(result){
    document.write("User clicked OK");
  } else {
    document.write("User clicked Cancel");
  }
</script>
```

prompt() – Input Dialog

```
<script>
  let name = window.prompt("Enter your name:");
  document.write("Hello " + name);
</script>
```

open() – Open New Window/Tab

```
<script>
  window.open("https://www.google.com");
</script>
```

close() – Close Window

```
<script>
  window.close();
</script>
```

setTimeout() – Execute After Delay

```
<script>
  setTimeout(function(){
    alert("This appears after 3 seconds");
  }, 3000);
</script>
```

setInterval() – Repeated Execution

```
<script>
  setInterval(function(){
    console.log("Running every 2 seconds");
  }, 2000);
</script>
```

history Object – Browser History

```
<script>
  window.history.back();    // Go to previous page
  window.history.forward(); // Go to next page
</script>
```

navigator Object – Browser Information

```
<script>
  document.write(navigator.appName);
  document.write("<br>");
  document.write(navigator.userAgent);
</script>
```

- window is the global object
- You can write alert() instead of window.alert()
- DOM manipulation is done through window.document
- The Window object is the top-level object in the DOM hierarchy and represents the browser window. It provides methods like alert(), prompt(), setTimeout() and properties like location, history, and navigator to control browser behavior using JavaScript.

Window Object Hierarchy

Window → Document → HTML → Head / Body → Elements

→ Location

→ History

→ Navigator

→ Screen

- window is the root object of the browser environment
- document controls the webpage content
- location manages URL navigation
- history manages browser navigation
- navigator gives browser & device info
- screen provides display details
- All JavaScript global variables & functions belong to window

DOM Manipulation (Document Object Model)

DOM represents the HTML page as a tree of objects.

JavaScript uses DOM methods to access and modify HTML elements dynamically.

getElementById()

Selects one element using its unique id.

```
<p id="demo">Hello</p>
<button onclick="changeText()">Click</button>
<script>
function changeText() {
    document.getElementById("demo").innerHTML = "Welcome to GEU";
}
</script>
```

getElementsByName()

Selects multiple elements with same name (used for radio/checkbox).

```
<input type="radio" name="gender" value="Male"> Male
<input type="radio" name="gender" value="Female"> Female
<button onclick="showGender()">Submit</button>
<script>
function showGender() {
   var g = document.getElementsByName("gender");
   for (var i = 0; i < g.length; i++) {
      if (g[i].checked) {
         alert(g[i].value);
      }
   }
}
</script>
```

getElementsByClassName()

Selects all elements of same class.

```
<p class="text">One</p>
<p class="text">Two</p>
<button onclick="changeAll()">Change</button>

<script>
function changeAll() {
    var x = document.getElementsByClassName("text");
    for (var i = 0; i < x.length; i++) {
        x[i].style.color = "red";
    }
}
</script>
```

querySelector()

Selects first matching element using CSS selector.

```
<p class="msg">Hello</p>
<button onclick="changeMsg()">Click</button>
<script>
function changeMsg() {
   document.querySelector(".msg").innerHTML = "DOM is Powerful";
}
</script>
```

JavaScript Events

Events occur due to user actions (click, type, move mouse).

<u>onclick</u>

```
<button onclick="sayHi()">Click Me</button>
<script>
function sayHi() {
   alert("Button Clicked");
}
</script>
```

Onchange

```
<select onchange="showCourse(this.value)">
  <option value="">Select</option>
  <option value="B.Tech">B.Tech</option>
  <option value="MCA">MCA</option>
</select>

<script>
function showCourse(val) {
    alert("Selected: " + val);
}
</script>
```

onmouseover

```
<p onmouseover="changeColor(this)">Hover Me</p>
<script>
function changeColor(x) {
   x.style.color = "blue";
}
</script>
```

onkeyup

```
<input type="text" onkeyup="showText(this.value)">
<p id="out"></p>
<script>
function showText(val) {
   document.getElementById("out").innerHTML = val;
}
</script>
```

Dynamic HTML

Dynamic HTML means changing content, style, and structure at runtime using JavaScript.

Change Text

```
<p id="t">Old Text</p>
<button onclick="change()">Change</button>
<script>
function change() {
    document.getElementById("t").innerHTML = "New Text";
}
</script>
```

Change Color

```
<p id="c">Color Me</p>
<button onclick="color()">Red</button>

<script>
function color() {
   document.getElementById("c").style.color = "red";
}
</script>
```

Change Visibility

```
<p id="p">Hide Me</p>
<button onclick="hide()">Hide</button>

<script>
function hide() {
   document.getElementById("p").style.display = "none";
}
</script>
```

Password Strength Checker

```
Password:
<input type="password" id="pwd" onkeyup="checkStrength()">
<p id="strength"></p>

<script>
function checkStrength() {
   let pwd = document.getElementById("pwd").value;
   let strength = "Weak";
   if (pwd.length >= 8 && /[A-Z]/.test(pwd) && /[0-9]/.test(pwd)) {
      strength = "Strong";
   } else if (pwd.length >= 6) {
      strength = "Medium";
   }
   document.getElementById("strength").innerHTML =
      "Password Strength: " + strength;
}
</script>
```

Show / Hide Password

Allows users to toggle password visibility for better usability.

```
Password:
<input type="password" id="pass">
<input type="checkbox" onclick="toggle()"> Show Password
<script>
function toggle() {
   let p = document.getElementById("pass");
   p.type = (p.type === "password") ? "text" : "password";
}
</script>
```

Character Counter in Textarea

Displays remaining or used characters while typing.

```
<textarea id="msg" rows="4" cols="30" onkeyup="countChar()"></textarea>
<p id="count">Characters: 0</p>
<script>
function countChar() {
    let text = document.getElementById("msg").value;
    document.getElementById("count").innerHTML =
        "Characters: " + text.length;
}
</script>
```

JAVASCRIPT FUNCTIONS & ARRAYS

Functions

A function is a reusable block of code that performs a specific task.

```
<script>
function greet() {
    alert("Welcome to JavaScript");
}
greet();
</script>
```

Parameterized Functions

A function that accepts parameters (inputs).

```
<script>
function greetUser(name) {
    alert("Hello " + name);
}
greetUser(“Amit");
</script>
```

Return Values

Functions can return a value using return.

```
<script>
function add(a, b) {
    return a + b;
}
let result = add(5, 3);
console.log(result);
</script>
```

Arrays

An array stores multiple values in a single variable.

```
<script>
let subjects = ["DBMS", "OS", "CN"];
console.log(subjects);
Document.write(subjects);
</script>
```

Array Methods

```
<script>
let arr = ["A", "B"];
arr.push("C");//push() – Add at end
console.log(arr);
</script>

<script>
arr.pop();//pop() – Remove last
console.log(arr);
</script>
```

shift() – Remove first

```
<script>
arr.shift();
console.log(arr);
</script>
```

join() – Convert array to string

```
<script>
let courses = ["B.Tech", "MCA", "MBA"];
console.log(courses.join(", "));
</script>
//length – Number of elements
<script>
console.log(courses.length);
</script>
```

STRING METHODS

//toUpperCase()

Converts string to uppercase.

```
<script>
let name = "graphic era";
console.log(name.toUpperCase());
</script>
```

//trim() Removes extra spaces from both ends.

```
<script>
let text = "   hello world   ";
console.log(text.trim());
</script>
```

substring()

Extracts part of a string.

```
<script>
let str = "JavaScript";
console.log(str.substring(0, 4)); // Java
</script>
```

Page Redirection

Page redirection is used to navigate the user to another page automatically or after an event using JavaScript.

```
<button onclick="redirect()">Go to Google</button>
```

```
<script>
function redirect() {
    window.location.href = "https://www.google.com";
}
</script>
```

Form Reset using JavaScript

Form reset clears all input fields and restores default values.

```
<form id="myForm">
   Name: <input type="text"><br><br>
   Email: <input type="email"><br><br>

   <button type="button" onclick="resetForm()">Reset</button>
</form>

<script>
function resetForm() {
   document.getElementById("myForm").reset();
}
</script>
```

JavaScript Objects & Classes

JavaScript Objects

- A JavaScript object is a real-world entity representation that stores related data (properties) and behaviour (methods) in the form of key–value pairs.
- Objects are used to model:
- Student, Employee, Product, User, etc.

Key Characteristics

- Properties → variables inside object
- Methods → functions inside object
- Accessed using **dot (.) operator**

```
<script>
let student = {
    name: “Amit",
    roll: 101,
    course: "B.Tech",
    display: function () {
        return this.name + " - " + this.course;
    }
};

console.log(student.name);
console.log(student.display());
</script>
```

Ways to Create Objects

- Using Object Literal

```
let obj = { key: value };
```

- Using Constructor Function

```
<script>
function Student(name, roll) {
   this.name = name;
   this.roll = roll;
}

let s1 = new Student("Amit", 102);
</script>
```

JavaScript Classes

A class is a blueprint or template used to create multiple objects with similar properties and methods.

- class keyword
- constructor() initializes object
- this refers to current object
- Methods defined without function keyword

```
<script>
class Student {
   constructor(name, course) {
      this.name = name;
      this.course = course;
   }
   info() {
      return this.name + " studies " + this.course;
   }
}
let st1 = new Student("Neha", "MCA");
console.log(st1.info());
</script>
```

Class Inheritance

```
<script>
class Person {
   constructor(name) {
      this.name = name;
   }
   greet() {
      return "Hello " + this.name;
   }
}
class Teacher extends Person {
   constructor(name, subject) {
      super(name);
      this.subject = subject;
   }

   details() {
      return this.greet() + ", Subject: " + this.subject;
   }
}
let t1 = new Teacher("Dr. Singh", "DBMS");
console.log(t1.details());
</script>
```

Full Stack vs MERN Stack

Full Stack (broad concept)

<u>A full stack developer works on both:</u>

- Frontend (UI): HTML, CSS, JavaScript + frameworks (React/Angular/Vue)
- Backend (server + APIs): Node.js / Java / Python / .NET / PHP, etc.
- Database: MySQL / PostgreSQL / MongoDB / Oracle, etc.
- So Full Stack = any combination of frontend + backend + database + deployment.

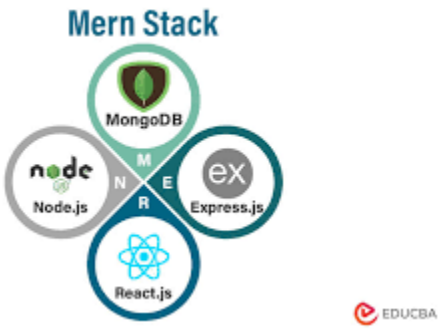
<u>Examples of full stack combos</u>

- React + Java Spring Boot + MySQL
- Angular + .NET + SQL Server
- Vue + Django + PostgreSQL
- React + Node + MongoDB (this one is MERN)

MERN Stack

MERN is a specific full-stack technology stack:

- **M** = MongoDB (database)
- **E** = Express.js (backend framework on Node)
- **R** = React (frontend)
- **N** = Node.js (runtime/backend)



So MERN ⊂ Full Stack (MERN is a subset of full stack).

Key differences

- **Scope**
  - Full Stack: wide, many tech options
  - MERN: fixed set (MongoDB + Express + React + Node)
- **Flexibility**
  - Full Stack: more flexible (you can mix any tools)
  - MERN: less flexible but very consistent as a package
- **Learning path**
  - Full Stack: can be broader depending on chosen tech
  - MERN: smoother learning path (all JavaScript-based)
- **Best for**
  - Full Stack: enterprise stacks, varied projects
  - MERN: modern web apps, startups, rapid development, JS-heavy teams

Regular Expression

A Regular Expression (RegEx) is a sequence of characters that defines a search pattern.
In JavaScript, regular expressions are mainly used for string validation, pattern matching, searching, and replacing text.

RegEx in JavaScript

In JavaScript, a regular expression can be created using:

Literal notation: /pattern/

RegExp object: new RegExp("pattern")

The most commonly used method is:

```
pattern.test(string)
```

Explanation of Password RegEx

/^(?=.*[a-z])(?=.*[A-Z])(?=.*\d)(?=.*[@$!%*?&]).{8,}$/

Meaning:

^ → Start of string
(?=.*[a-z]) → At least one lowercase letter
(?=.*[A-Z]) → At least one uppercase letter
(?=.*\d) → At least one digit
(?=.*[@$!%*?&]) → At least one special character
{8,} → Minimum 8 characters
$ → End of string

Password At least 6 characters, letters and numbers:

```
let simplePasswordPattern = /^(?=.*[A-Za-z])(?=.*\d)[A-Za-z\d]{6,}$/;
let pass = "abc123";
console.log(simplePasswordPattern.test(pass)); // true
```

URL Validation

```
let urlPattern = /^(https?:\/\/)?([\w\-])+\.{1}[a-zA-Z]{2,}(\/\S*)?$/;
let url = "https://www.geu.ac.in";
console.log(urlPattern.test(url)); // true
```

Date Format (DD/MM/YYYY)

```
let datePattern = /^(0[1-9]|[12][0-9]|3[01])\/(0[1-9]|1[0-2])\/\d{4}$/;
let date = "29/01/2026";
console.log(datePattern.test(date)); // true
```

Only Numbers

```
let numberPattern = /^\d+$/;
let value = "12345";
console.log(numberPattern.test(value)); // true
```

PIN Code – 6 Digits

```
let pinPattern = /^\d{6}$/;
let pincode = "248001";
console.log(pinPattern.test(pincode)); // true
```

Email Address Validation

```
let emailPattern = /^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$/;
let email = "user@example.com";
console.log(emailPattern.test(email)); // true
```

Password Validation

```
let passwordPattern =/^(?=.*[a-z])(?=.*[A-Z])(?=.*\d)(?=.*[@$!%*?&])[A-Za-z\d@$!%*?&]{8,}$/;
let password = "Admin@123";
console.log(passwordPattern.test(password)); // true
```

JSON

JSON (JavaScript Object Notation) is a lightweight, text-based data interchange format used to store and exchange data between a client (browser/JavaScript) and a server.

Although it is derived from JavaScript object syntax, JSON is language-independent, meaning it works with Java, Python, PHP, C#, etc.

<u>JSON vs JavaScript</u>

JavaScript Object	JSON
{ name: "Amit", age: 40 }	{ "name": "Amit", "age": 40 }
Keys can be unquoted	Keys must be in double quotes
Can store functions	✕ No functions
Used inside JS	Used for data exchange

Characteristics of JSON

- Text-based format: Written in plain text, making it platform-independent.
- Language-independent: Although derived from JavaScript syntax, JSON is supported by almost all programming languages.
- Lightweight: Uses minimal syntax compared to XML, resulting in faster data transmission.
- Structured data representation: Organizes data in a clear key–value format.

JSON Data Structure

JSON represents data using two main structures:

<u>Objects</u>

- Unordered collections of key–value pairs
- Keys are strings; values can be strings, numbers, objects, arrays, booleans, or null

<u>Arrays</u>

- Ordered lists of values
- Values can be of any valid JSON data type

JSON Data Types

- String – Text data enclosed in double quotes
- Number – Integer or floating-point values
- Boolean – true or false
- Array – Ordered collection of values
- Object – Collection of key–value pairs
- Null – Represents an empty or unknown value

Why JSON is Used

- Data exchange between client and server (e.g., web applications, APIs)
- Configuration files and data storage
- Interoperability between different systems and platforms
- Efficient alternative to XML due to simpler syntax and smaller size

JSON vs XML

Feature	JSON	XML
Readability	Easy	Verbose
Data Size	Compact	Larger
Parsing Speed	Fast	Slower
Data Structure	Key–Value	Tag-based

JSON is a standard data representation format designed for efficient data exchange. Its simplicity, readability, and cross-platform support make it the backbone of modern web services, APIs, and client-server communication.

but No Support for comments, limited data types and Not suitable for complex document markup.

Standard Format for Data Exchange

- JSON is the default format for APIs
- Used in AJAX, Fetch API, REST APIs
- Almost all modern web applications use JSON

```
{
  "id": 101,
  "name": "Laptop",
  "price": 55000
}
```

Easy to Convert Between JSON and JavaScript Objects

JavaScript provides built-in methods:

- JSON.parse() → JSON string → JS object
- JSON.stringify() → JS object → JSON string

JSON vs JavaScript Object

<u>JavaScript Object//</u>JS Object → used for **programming logic**

```
let person = {
  name: “Amit",
  age: 35,
  department: "CSE"
};
```

<u>JSON //</u>JSON → used for **data transfer**

```
{
  "name": “Amit",
  "age": 35,
  "department": "CSE"
}
```

Converting JSON to JavaScript Object

```
let jsonData = '{"name":“Amit","age":35}';
let obj = JSON.parse(jsonData);
console.log(obj.name); // Amit
console.log(obj.age);  // 35
```

JSON.parse() is used to convert a JSON-formatted string into a JavaScript object, so that the data can be accessed, manipulated, and processed within a JavaScript program.

SON data received from external sources (server, API, file, localStorage) is always in string format.

JavaScript cannot directly work with JSON strings as objects—so we use JSON.parse().

Converting JavaScript Object to JSON

```
let student = {
  roll: 101,
  name: "Amit",
  branch: "CSE"
};
let jsonString = JSON.stringify(student);
console.log(jsonString);
```

//JSON.stringify() is used to convert a JavaScript object (or array) into a JSON-formatted string, so that it can be stored, transmitted, or sent over a network.

JSON with Array

```
{
  "students": [
    { "roll": 1, "name": "Amit" },
    { "roll": 2, "name": "Riya" },
    { "roll": 3, "name": "Neha" }
  ]
}
```

Accessing in JavaScript

```
let data = {
  students: [
    { roll: 1, name: "Amit" },
    { roll: 2, name: "Riya" },
    { roll: 3, name: "Neha" }
  ]
};
console.log(data.students[1].name); // Riya
```

JSON Example with HTML

```
<!DOCTYPE html><html><body>
<h3>Student Details</h3>
<p id="output"></p>
<script>
let jsonData = '{"name":“Amit","designation":"Professor","dept":"CSE"}';
let obj = JSON.parse(jsonData);
document.getElementById("output").innerHTML =
  "Name: " + obj.name + "<br>" +
  "Designation: " + obj.designation + "<br>" +
  "Department: " + obj.dept;
</script></body></html>
```

Practical Example: Student Registration + Save to JSON + Read back

```
!DOCTYPE html>
<html>
<head>
  <title>JSON Practical Example</title>
  <style>
    body { font-family: Arial; margin: 20px; }
    input, button { padding: 8px; margin: 5px; }
    table { border-collapse: collapse; width: 60%; margin-top: 10px; }
    th, td { border: 1px solid #999; padding: 8px; text-align: left; }
  </style>
</head>
<body>
```

```
<h2>Student Registration (JSON Practical)</h2>
<input id="name" placeholder="Enter Name">
<input id="roll" placeholder="Enter Roll No">
<input id="dept" placeholder="Enter Department">
<button onclick="saveStudent()">Save Student</button>
<button onclick="loadStudents()">Load Students</button>
<h3>Saved Students</h3>
<table>  <thead>
    <tr>  <th>Roll</th>
      <th>Name</th>
      <th>Department</th> </tr>
  </thead>
  <tbody id="tableBody"></tbody>
</table>
```

```
<script>
function saveStudent() {
  // 1) Read values from inputs
  let name = document.getElementById("name").value.trim();
  let roll = document.getElementById("roll").value.trim();
  let dept = document.getElementById("dept").value.trim();
  if (!name || !roll || !dept) {
    alert("Please fill all fields!");
    return;
  }
  // 2) Create a JavaScript object (NOT JSON yet)
  let studentObj = {
    roll: roll,
    name: name,
    dept: dept
  };
```

```
// 3) Get existing students from localStorage (it is JSON string)
  let oldData = localStorage.getItem("students"); // string OR null

  // 4) Convert JSON string to JS array
  let studentsArray = oldData ? JSON.parse(oldData) : [];

  // 5) Add new student object into array
  studentsArray.push(studentObj);

  // 6) Convert JS array -> JSON string (because storage needs string)
  localStorage.setItem("students", JSON.stringify(studentsArray));
```

```
  // clear input
   document.getElementById("name").value = "";
   document.getElementById("roll").value = "";
   document.getElementById("dept").value = "";

   alert("Student saved in JSON format!");
   loadStudents(); // refresh table
}
```

```
function loadStudents() {
  let data = localStorage.getItem("students"); // JSON string

  // If no data, show empty
  if (!data) {
    document.getElementById("tableBody").innerHTML = "<tr><td
colspan='3'>No students found</td></tr>";
    return;
  }
```

```
// Convert JSON string back into JS array of objects
 let studentsArray = JSON.parse(data);
 let rows = "";
 studentsArray.forEach(s => {
   rows += `
     <tr>
       <td>${s.roll}</td>
       <td>${s.name}</td>
       <td>${s.dept}</td>
     </tr>
   `;
 });
 document.getElementById("tableBody").innerHTML = rows;
}
```

```
// Auto-load table on page refresh
loadStudents();
</script>
</body>
</html>
```

In this program:

- We take user input and create a JavaScript object.
- We store data in localStorage, but localStorage can store only STRING.
- So we convert object/array into JSON string using JSON.stringify()
- When reading back, we convert JSON string into object using JSON.parse()

Event Listener in JavaScript

What is an Event?

An **event** is an action that occurs in the browser, such as:

- Clicking a button
- Moving the mouse
- Typing on the keyboard
- Submitting a form
- Page loading

What is an Event Listener?

An event listener is a JavaScript mechanism that:

- listens for a specific event on an element and executes a function when that event occurs.

It is added using:

```
element.addEventListener(event, function);
```

Syntax of addEventListener()

```
element.addEventListener("event", callbackFunction);
```

Components:

- event → type of event (e.g., "click", "mouseover")
- callbackFunction → function executed when event occurs

COMMON JAVASCRIPT EVENTS

Event	Triggered When
Click	Mouse click
dblclick	Double click
Mouseover	Mouse enter element
Mouseout	Mouse leaves element
Keydown	Key is pressed
keyup	Key is released
Change	Input values chages
Submit	Form is submitted
load	Page finishing loading

Event Object (event or e)

When an event occurs, JavaScript creates an event object.

```
element.addEventListener("click", function(e) {
  console.log(e.target);
});
```

Common Event Object Properties:

e.target → element that triggered the event

e.type → event type

e.preventDefault() → stops default action

e.stopPropagation() → stops bubbling

Button Click

```
<button id="btn">Click Me</button>


<script>
  document.getElementById("btn").addEventListener("click", function ()
{
    alert("Button clicked using Event Listener");
  });
</script>
```

Mouse Over & Mouse Out

```
<div id="box" style="width:150px;height:150px;border:2px solid black;">
  Hover Me
</div>
<script>
  const box = document.getElementById("box");
  box.addEventListener("mouseover", () => {
    box.style.background = "lightgreen";
  });
  box.addEventListener("mouseout", () => {
    box.style.background = "white";
  });
</script>
```

Keyboard Event

```
<input type="text" id="name" placeholder="Type here">

<script>
  document.getElementById("name").addEventListener("keyup",
function (e) {
    console.log("Key pressed:", e.key);
  });
</script>
```

Form Submit with preventDefault()

```
<form id="myForm">
  <input type="text" required>
  <button type="submit">Submit</button>
</form>

<script>
  document.getElementById("myForm").addEventListener("submit", function (e) {
    e.preventDefault();
    alert("Form submission prevented!");
  });
</script>
```

Multiple Events on Same Element

```
const btn = document.getElementById("btn");

btn.addEventListener("click", () => console.log("Clicked"));
btn.addEventListener("mouseover", () => console.log("Hovered"));
```

CASCADING STYLE SHEETS (CSS)

Need for CSS

CSS (Cascading Style Sheets) is used to describe the presentation, formatting, and layout of HTML documents. It controls how web content appears on different devices.

CSS:

- Improves code readability
- Reduces HTML size
- Enhances website performance
- Enables responsive web design
- Easy to update entire website from one file

Example:

```
<h1 style="color: red;">Welcome</h1>
```

CSS stands for Cascading Style Sheets.
It defines how HTML elements should be displayed in terms of:

- Color
- Font
- Spacing
- Layout
- Animation

Characteristics

- CSS rules follow a cascade priority
- Supports multiple style sheets
- Compatible with all modern browsers

Basic CSS Syntax and Structure

```
selector {
    property: value;
            }
```

- Selector – HTML element to style
- Property – Styling feature
- Value – Assigned value

Example:

```
p {
    color: blue;
    font-size: 16px;
}
```

Multiple Properties

```
h1 {
   color: green;
   text-align: center;
   text-transform: uppercase;
    }
```

Using CSS in Web Pages

CSS controls:

- Text appearance
- Backgrounds
- Layout
- Positioning
- Animations
- Responsive behavior

```
body {
    background-color: lightblue;
    font-family: Arial;
        }
```

Types of CSS

Inline CSS

- Written inside HTML elements
- Overrides other CSS types

Example:

```
<p style="color:red;">Inline CSS</p>
```

Disadvantage:

- Hard to maintain
- Repetition of code

Internal CSS:

- Defined inside <style> tag
- Applies to single page

```
<style>
p { color: blue; }
</style>
```

External CSS:

- Stored in .css file
- Reusable and maintainable

```
<link rel="stylesheet" href="style.css">
```

Background Image, Color and Properties

Background Color:

```
body {
    background-color: #f4f4f4;
      }
```

Background Image:

```
body {
    background-image: url("bg.jpg");
    background-repeat: no-repeat;
    background-position: center;
    background-size: cover;
}
```

Using Fonts in CSS

Font Properties

```
p {
    font-family: "Times New Roman", serif;
    font-size: 18px;
    font-style: italic;
    font-weight: bold;
}
```

Borders and CSS Box Model

```
div {
   border: 3px solid black;
    }
```

Border Styles:

- solid
- dashed
- dotted
- double

CSS Box Model Components

- Content
- Padding
- Border
- Margin

```
div {
    width: 300px;
    padding: 20px;
    border: 5px solid red;
    margin: 15px;
}
```

Introduction to Bootstrap

Bootstrap is a free, open-source CSS framework for developing responsive websites quickly.

Features:

- Mobile-first design
- 12-column grid system
- Ready-made components
- Browser compatibility

```
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
<div class="container">
    <button class="btn btn-success">Submit</button>
</div>
```

What is Bootstrap Grid System?

The Bootstrap Grid System is a responsive, mobile-first layout system provided by Bootstrap that allows developers to divide a web page into rows and columns and arrange content neatly across different screen sizes (mobile, tablet, desktop).

The Bootstrap grid system uses a 12-column layout to design flexible and responsive web pages.
It is based on containers, rows, and columns.

<u>Basic Structure:</u>

```
Container
 └── Row
     └── Columns (total = 12)
```

Why 12 Columns?

- 12 is easily divisible by 1, 2, 3, 4, 6
- Makes layout design flexible
- Supports equal and unequal column layouts

Visual Diagram (12-Column Grid):

```
|1|2|3|4|5|6|7|8|9|10|11|12|
```

Example Layouts:

```
Two columns:      |------6------|------6------|
Three columns:    |----4----|----4----|----4----|
Sidebar layout:   |----4----|--------8--------|
```

Main Components

Container:

Defines the width of the page.

- .container → fixed width
- .container-fluid → full width

```
<div class="container">
```

Row:

Creates a horizontal group of columns.

```
<div class="row">
```

Columns:

Specify how many columns an element occupies (out of 12).

```
<div class="col-md-6">
```

```
<div class="container">
  <div class="row">
    <div class="col-md-6">Left</div>
    <div class="col-md-6">Right</div>
  </div>
</div>
```

Behavior:

- On desktop → two columns side by side
- On mobile → columns stack vertically

Introduction to XML

XML (eXtensible Markup Language) is a text-based markup language used to store and transport data in a structured, readable format. It is platform independent and designed to be self-descriptive, meaning the tags describe the data they contain.

Key points

- XML is not a programming language.
- XML is not meant for displaying data (HTML is for display).
- XML is mainly for data representation + data exchange.
- XML allows users to create custom tags.

Uses of XML

XML is used when data must be:

- Structured (hierarchical)
- Readable and portable
- Easy to share between systems

Common uses

- Data exchange between applications
  - Example: sharing data between a Java app and a PHP app.
- Web services (SOAP)
  - SOAP messages are XML-based.
- Configuration files
  - Many software tools use XML configuration.
- Storing semi-structured data
  - Example: product catalogs, invoices, library records.
- Document formats
  - Examples: Office formats like DOCX, XLSX are zipped XML-based formats.
- Metadata and feeds
  - Some publishing standards use XML (e.g., RSS uses XML format).

SOAP (Simple Object Access Protocol) is an XML-based messaging protocol for exchanging structured information between applications over a network, enabling communication between diverse systems regardless of their underlying programming languages or platforms.

Message Format:

All SOAP messages are fundamentally standard XML documents. The XML structure provides a standardized, machine-parsable format for the message payload.

Structure Definition:

SOAP uses XML Schemas (XSD) to define the precise structure, elements, and data types that can appear in a SOAP message, ensuring a rigid contract for communication.

Interoperability:

By leveraging XML's platform-independent nature, SOAP enables applications built on different technologies (e.g., a C# application and a Java application) to "understand" and process the same message format.

Structure of a SOAP Message

A SOAP message is an XML document composed of specific, standardized elements:

<Envelope>:

The mandatory root element that identifies the XML document as a SOAP message and defines the overall framework for processing it.

<Header>:

An optional element that contains application-specific control information, such as authentication credentials, transaction IDs, or routing details. This information can be processed by intermediary applications along the message path.

<Body>:

The mandatory element that contains the actual message payload, which includes the request details for a remote procedure call (RPC) or the response data.

<Fault>:

An optional element that, if present, appears within the <Body> and is used to report error messages and status information if a request fails.

SimpleXML

SimpleXML usually refers to the PHP extension that provides a simple way to read and work with XML data. It converts XML into an object-like structure so you can access elements easily.

```
<students>
  <student>
    <name>Amit</name>
    <marks>85</marks>
  </student>
  <student>
    <name>Neha</name>
    <marks>90</marks>
  </student>
</students>
```

Access elements using PHP

```
<?php
$xml = simplexml_load_file("students.xml");

foreach($xml->student as $s){
  echo $s->name . " - " . $s->marks . "<br>";
}
?>
```

XML Key Components

XML Declaration

<?xml version="1.0" encoding="UTF-8"?>

Elements (Tags)

Building blocks of XML:

<name>Amit</name>

Attributes

Extra information inside a tag:

<student id="101">Amit</student>

Root Element

Every XML must have exactly one root element:

<students> ... </students>

DTD (Document Type Definition)

DTD defines the structure and legal elements of an XML document.

Purpose of DTD

Validates XML structure

Ensures correct element order

Defines allowed elements and attributes

Internal DTD Example

```
<!DOCTYPE student [
<!ELEMENT student (name, course)>
<!ELEMENT name (#PCDATA)>
<!ELEMENT course (#PCDATA)>
]>
<student>
   <name>Amit</name>
   <course>CSE</course>
</student>
```

Limitations of DTD

- No data types (only text)
- Not written in XML syntax
- Less powerful

XML Schema (XSD)

XML Schema (XSD – XML Schema Definition) is an advanced way to define XML structure with data types and constraints.

Advantages over DTD

- Written in XML
- Supports data types (int, string, date)
- Supports constraints and validation rules
- Namespace support

XSD Example

XML file

```
<student>
  <name>Amit</name>
  <age>30</age>
</student>
```

Schema file (student.xsd)

```
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema">
 <xs:element name="student">
  <xs:complexType>
   <xs:sequence>
    <xs:element name="name" type="xs:string"/>
    <xs:element name="age" type="xs:integer"/>
   </xs:sequence>
  </xs:complexType>
 </xs:element>
</xs:schema>
```

jQuery

jQuery is a fast, lightweight, and feature-rich JavaScript library designed to simplify:

- HTML DOM manipulation
- Event handling
- Animations
- AJAX operations
- Cross-browser compatibility

It follows the principle:

"Write less, do more."

Why jQuery?

Before jQuery, JavaScript required long code for simple tasks.

Pure JavaScript

```
document.getElementById("demo").style.color = "red";
```

Same in jQuery:

```
$("#demo").css("color", "red");
```

Installing and Using jQuery Library

There are two ways to use jQuery:

1. Using CDN: Add this inside <head> or before </body>:

   <script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>

   No download required

   Faster loading (cached globally)

2. Download and Use Locally

   Download from: https://jquery.com

   Save jquery.min.js in your project folder

   Add:

   <script src="jquery.min.js"></script>

What is a CDN?

A Content Delivery Network (CDN) is a distributed network of servers located in different geographical locations that work together to deliver web content (like images, videos, CSS, JavaScript, APIs, etc.) to users faster and more reliably.

Basic Structure of jQuery

```
<html><head><title>jQuery Example</title>
<script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>
<script>
$(document).ready(function(){
   $("#btn").click(function(){
      $("#demo").text("Hello Students!");
   });
});
</script> </head><body>
<p id="demo">Original Text</p>
<button id="btn">Click Me</button>
</body>
</html>
```

jQuery Syntax

$(selector).action();

Part	Meaning
$	jQuery symbol
selector	Selects HTML element
action()	Performs action

Common Selectors:

Selector	Meaning
$("p")	Select all paragraphs
$("#id")	Select element by ID
$(".class")	Select elements by class
$("div p")	Select p inside div

Common Actions

Method	Purpose
.hide()	Hides selected element
.show()	Shows hidden element
.toggle()	Switch between hide & show
.css()	Changes CSS properties dynamically
.text()	Changes only text (no HTML formatting)
.html()	Changes content including HTML tags
.val()	Gets/sets input field value
.addClass()	Adds a CSS class
.removeClass()	Removes a CSS class